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CODE A

Mock JEE Main - 10 (CBT) | JEE - 2024

Date: 28/03/2024 **Maximum Marks: 300**

Timing: 3:30 PM to 6:30 PM **Duration : 3.0 Hours**

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **300.**
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices **(A)**, **(B)**, **(C)** and **(D)**, out of which **ONLY ONE CHOICE** is correct.
4. **Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted. You will NOT be allowed to attempt the sixth question. If you wish to attempt any other question apart from the five already attempted, then you will have to delete any one response from the five previously answered and then proceed to answer the new one.**
   The answer to each question should be **rounded off to the nearest integer**.
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them**.

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.

Name of the Candidate (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ Invigilator's Signature _____

PART I: PHYSICS	MARKS: 100

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. A galvanometer has a resistance of 20Ω and reads full-scale when $0.2V$ is applied across it. To convert it into a $10A$ ammeter, the galvanometer coil should have a:

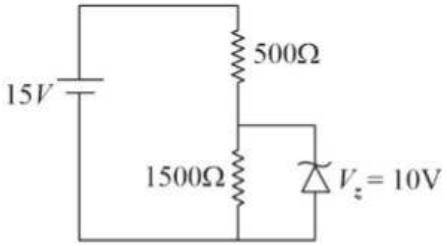
(A) 0.01Ω resistor connected parallel to it

(B) 0.02Ω resistor connected parallel to it

(C) 200Ω resistor connected in series with it

(D) 2000Ω resistor connected in series with it

2. Find power through Zener diode in voltage regulator circuit shown in the figure:



(A) $100\times10^{-3}W$ **(B)** $50\times10^{-3}W$ **(C)** $500\times10^{-3}W$ **(D)** $33.3\times10^{-3}W$

3. What is the value of inductance 'L' for which the current is maximum in a series LCR circuit with $C = 10\mu F$ and $\omega = 1000s^{-1}$?

(A) 1mH **(B)** 10 mH **(C)** 100mH **(D)** 1000 mH

4. The dimension formula of planck's constant (h) is:

(A) $[ML^2T^{-1}]$ **(B)** $[ML^{-2}T^{-1}]$ **(C)** $[ML^2T^{-2}]$ **(D)** $[MLT^{-1}]$

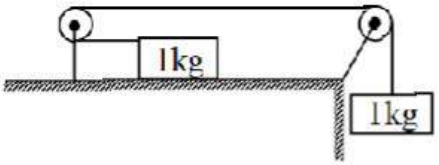
5. If a man at the equator would weight $\left(\frac{3}{5}\right)$th of his weight, the angular speed of the earth is:

(A) $\sqrt{\frac{2g}{5R}}$ **(B)** $\sqrt{\frac{g}{R}}$ **(C)** $\sqrt{\frac{R}{g}}$ **(D)** $\sqrt{\frac{2}{5}\frac{R}{g}}$

6. The ratio for the speed of the electron in the 3rd orbit of He^+ to the speed of the electron in the 3rd orbit of hydrogen atom will be:

(A) 1:1 **(B)** 1:2 **(C)** 4:1 **(D)** 2:1

7. A block of mass 1 kg is placed on a rough horizontal surface connected by a light string passing over two smooth pulleys as shown. Another block of $1Kg$ is connected to the other end of the string. The acceleration of the system is: [Coefficient of friction $\mu = 0.2$]



(A) $0.8\,g$ **(B)** $0.4\,g$ **(C)** $0.5\,g$ **(D)** Zero

8. A stone of mass $1.5Kg$ is tied at the end of $0.5m$ long string and whirled in a vertical circular path at constant speed of $2ms^{-1}$. The maximum tension in the string is: $[g = 10ms^{-2}]$

(A) $27\,N$ **(B)** $3\,N$ **(C)** $90\,N$ **(D)** $15\,N$

9. Two moles of a monoatomic gas is mixed with 3 moles of a diatomic gas. Find $\frac{C_P}{C_V}$ of mixture of gases.

(A) $\frac{11}{21}$ **(B)** $\frac{21}{11}$ **(C)** $\frac{21}{31}$ **(D)** $\frac{31}{21}$

10. A series combination of R, L and C is connected to an A.C. source. If the resistance is 3Ω and reactance is 4Ω. The power factor of the circuit is:

(A) 0.4 **(B)** 0.6 **(C)** 0.8 **(D)** 1.0

11. The energy provided to an Li^{++} atom in ground state so that it can emit only three different photons will be:

(A) $13.6\,eV$ **(B)** $91.8\,eV$ **(C)** $108.8\,eV$ **(D)** $12.1\,eV$

12. The pressure and volume of 1 mole ideal gas are related as $PV^{3/4} =$ constant. The work done by gas when its temperature is increased from T_0 to $3T_0$ is:

(A) $4RT_0$ **(B)** $8RT_0$ **(C)** $16RT_0$ **(D)** $32RT_0$

13. The length (l) and acceleration due to gravity (g) was measured in the laboratory as $l = (5 \pm 0.05)\,m$, $g = (10 \pm 0.20)\,m/s^2$. The percentage error in time period of simple pendulum is:

(A) 0.5% **(B)** 1% **(C)** 1.5% **(D)** 2%

14. A particle is projected with a velocity v such that its range on the horizontal plane is twice the greatest height attained by it. The range of the projectile is (where g is acceleration due to gravity)

(A) $\frac{4v^2}{5g}$ **(B)** $\frac{4g}{5v^2}$ **(C)** $\frac{v^2}{g}$ **(D)** $\frac{4v^2}{\sqrt{5}\,g}$

15. A thin sheet of glass $(\mu = 1.5)$ of thickness 6 microns introduced in the path of one of interfering beams of a double slit experiment shifts the central fringe to a position previously occupied by fifth bright fringe. Then the wavelength of the light used is:

(A) 6000 Å **(B)** 3000 Å **(C)** 4500 Å **(D)** 7500 Å

16. The pitch of a screw gauge is 1 mm and there are 200 divisions on the circular scale. The least count of the screw will be:

(A) $0.5\mu m$ **(B)** $1\mu m$ **(C)** $5\mu m$ **(D)** $10\mu m$

17. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Clothes containing oil or grease stains cannot be cleaned by water wash.

Reason (R) : Because the angle of contact between the oil/grease and water is obtuse.

In the light of the above statements, choose the correct answer from the option given below:

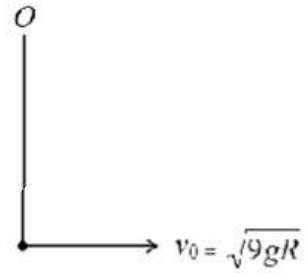
(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(C) (A) is true but (R) is false

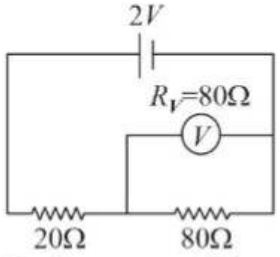
(D) (A) is false but (R) is true

18. A ball is projected with horizontal velocity $\upsilon_0 = \sqrt{9gR}$ at the bottom most point attached with inextensible string of length R and fixed at O as shown. Tension in the string in horizontal position:



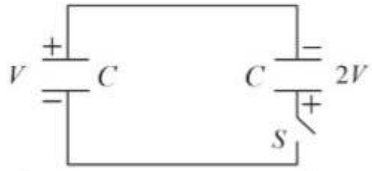
(A) $3\,mg$ **(B)** $5\,mg$ **(C)** $7\,mg$ **(D)** $9\,mg$

19. The reading of voltmeter in the following circuit will be:



(A) 2 volt **(B)** 0.80 volt **(C)** 1.33 volt **(D)** 1.60 volt

20. Two capacitor of equal capacitance are connected according to figure shown below. Now switch is closed. The decrease in energy of the combined system is:



(A) $\frac{3}{4}CV^2$ **(B)** $\frac{1}{4}CV^2$ **(C)** $2CV^2$ **(D)** $\frac{9}{4}CV^2$

SPACE FOR ROUGH WORK

SECTION-2

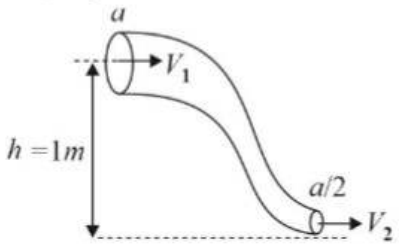
Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted.
The answer to each question should be **rounded off to the nearest integer**.

1. A particle is moving in one dimension (along x-axis) under the action of variable force. The variation of its position (x) with time (t) is given as $x = \frac{t^3}{3} - 2t^2 + 4t + 3$, where x is in m and t is in sec. The acceleration of the particle when its velocity becomes zero is _____ m/s^2.

2. A source of frequency 10 kHz when vibrated over the mouth of a closed organ pipe is in unison at 250K temperature. The beats produced when temperatures rises by 1 K is _____ Hz.

3. An ideal fluid of density 800 $kg\ m^{-3}$, flows smoothly through a bent pipe (as shown in the figure) that tapers in cross-sectional area from a to $\frac{a}{2}$. The pressure difference between the wide and narrow sections of pipe is 4100 Pa. At wider section, the velocity of fluid is $\frac{\sqrt{x}}{6} ms^{-1}$ for x = _____.

[Given: $g = 10 m/s^2$]



4. A square is formed by bending a wire of length 4π meter. If an electric current of $\sqrt{2}\pi A$ is flowing through the sides of square, the magnetic field at the centre of the polygon would be $x \times 10^{-7} T$. The value of x is _____.

5. In an electrical circuit, a battery is connected to pass 20C of charge through it in a certain given time. The potential difference between two plates of the battery is maintained at 15V. The work done by the battery is _____ J.

6. Focal length of a convex lens of refractive index 1.5 is 2 cm. Focal length of lens when immersed in a liquid of refractive index of 1.25 will be _____ cm.

7. The position of the centre of mass of a uniform semi-circular wire of radius 'R' placed in x-y plane with its centre at the origin and the line joining its ends as x-axis is given by $\left(0, \frac{xR}{\pi}\right)$. Then, the value of $|x|$ is _____.

8. Flux ϕ (in weber) in a closed circuit of resistance 10Ω varies with time 't' (in seconds) according to the equation $\phi = 6t^2 - 4t + 1$. The magnitude of the induced current in the circuit at t = 2 is _____ Ampere.

9. When light of frequency twice the threshold frequency is incident on the metal plate, the maximum velocity of emitted electron is v_1. When the frequency of incident radiation is increased to five times the threshold value, the maximum velocity of emitted electron becomes v_2. If $v_2 = xv_1$, the value of x will be _____.

10. Two identical conducting spheres with negligible volume have 2.1 nC and –0.1 nC charges, respectively. They are brought into contact and then separated by a distance of 0.5 m. The electrostatic force acting between the spheres is _____ $\times 10^{-9}$ N.

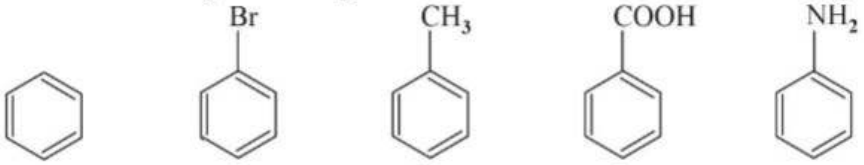
SPACE FOR ROUGH WORK

PART II: CHEMISTRY	MARKS: 100

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Correct order of reactivity in electrophilic substitution reaction of the following compounds?



i ii iii iv v

(A) i > ii > iii > iv > v **(B)** v > iii > ii > iv > i

(C) v > iii > i > ii > iv **(D)** iii > i > v > ii > iv

2. For atoms/ions of Isoelectronic series, which of the following is same for all atoms/Ions:

(A) Size **(B)** Number of protons and electrons

(C) Nuclear charge **(D)** Principal quantum number

3. Which of the following reactions are comproportionation reactions?

(i) $2H_2S(g) + SO_2(g) \rightarrow 3S(g) + 2H_2O(g)$

(ii) $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

(iii) $2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$

(iv) $Ag^{2+}(aq) + Ag(s) \rightarrow 2Ag^+(g)$

Choose the correct answer from the options given below:

(A) (ii), (iii), (iv) **(B)** (i), (ii), (iii) **(C)** (i), (ii) **(D)** (i), (iv)

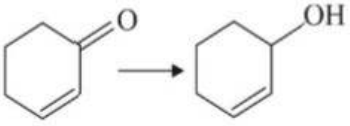
4. We have three aqueous solutions of 0.2 M Urea (A), 0.1 M NaCl (B), 0.01 M $MgCl_2$ (C) respectively.

The value of van't Hoff factor (i) for these solutions will be in the order:

[Consider NaCl, $MgCl_2$ are strong electrolyte]

(A) $i_A < i_C < i_B$ **(B)** $i_A < i_B < i_C$ **(C)** $i_A > i_B > i_C$ **(D)** $i_A = i_B = i_C$

5. Match List-I with List-II.

	List-I (Reactions)		List-II (Reagents)
(a)	$CH_3(CH_2)_5 - CN \rightarrow CH_3(CH_2)_5CHO$	**(i)**	$LiAlH_4$ / ether
(b)	$C_6H_5COC_6H_5 \rightarrow C_6H_5CH_2C_6H_5$	**(ii)**	$NH_2 - NH_2$ / KOH
(c)		**(iii)**	$NaBH_4$, H^+
(d)	$CH_3COCH_2COOH \rightarrow CH_3CH(OH)CH_2COOH$	**(iv)**	DIBAL – H, H_2O

Choose the correct answer from the options given below:

(A) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii) **(B)** (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

(C) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii) **(D)** (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

6. Reaction intermediate formed due to heterolytic bond cleavage are:

(A) Free radical and carbene **(B)** Cabocation and carbanion

(C) Free radical and carbanion **(D)** Cation only

7. In acidic medium, $KMnO_4$ shows oxidising action as represented in the half reaction:

$$MnO_4^- + XH^+ + Ye^- \rightarrow A + ZH_2O$$

X, Y, Z and A respectively are:

(A) 8, 4, 5 and MnO_2 **(B)** 4, 3, 2 and MnO_2

(C) 8, 4, 5 and Mn^{2+} **(D)** 8, 5, 4 and Mn^{2+}

8. Piston moves in an insulated cylinder from 10 L to 20 L against 2 atm pressure. The work done during this expansion is (in atm Ltr)

(A) –20 **(B)** 10 **(C)** +20 **(D)** –10

9. Given below are two statements. One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Haloalkanes react with KNO_2 to form alkyl nitrite as major product while with $AgNO_2$ form Nitroalkane as the major product.

Reason (R) : KNO_2 is ionic while $AgNO_2$ is covalent compound.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Assertion and Reason are correct and Reason is the correct explanation of assertion.

(B) Assertion is not correct but Reason is correct.

(C) Both Assertion and Reason are correct but Reason is not the correct explanation of assertion.

(D) Assertion is correct but Reason is not correct.

10. Given below are two statements:

Statement-I : A solution having $[Co(NH_3)_6]^{2+}$ ion is coloured.

Statement-II : A solution having $[Cu(NH_3)_4]^{2+}$ ion is colourless.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statement-I and Statement-II are correct

(B) Both Statement-I and Statement-II are incorrect

(C) Statement-I is correct but Statement-II are incorrect

(D) Statement-I is incorrect but Statement-II are correct

11. Given below are two statements:

Statement-I : Oxalic acid is a primary standard solution used for standardization of sodium hydroxide solution.

Statement-II : In this titration methyl orange can be used as indicator.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statement-I and Statement-II are incorrect.

(B) Both Statement-I is incorrect and Statement-II are correct.

(C) Statement-I is correct but Statement-II are incorrect.

(D) Both Statement-I and Statement-II are correct.

12. Given below are two statements:

Statement-I : Benzene can be used as a solvent for the Friedel-craft alkylation of bromobenzene.

Statement-II : Phenol does not undergo Friedel-Craft's reaction (alkylation and acylation)

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statement-I and Statement-II are correct

(B) Statement-I is incorrect but Statement-II is correct

(C) Statement-I is correct but Statement-II are incorrect

(D) Both Statement-I and Statement-II are incorrect

13. Which of the following complex is not homoleptic?

(A) $[Fe(NH_3)_4Cl_2]^+$ **(B)** $[Ni(H_2O)_6]^{2+}$

(C) $[Ni(CO)_4]$ **(D)** $[Cu(NH_3)_4]^{2+}$

14. Given below are two statements:

Statement-I : Carbolic acid and phenol are same organic compounds.

Statement-II : Carbolic acid and phenol have different functional groups.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Statement-I is correct but Statement-II is incorrect

(B) Both Statement-I and Statement-II are incorrect

(C) Statement-I is incorrect but Statement-II are correct

(D) Both Statement-I and Statement-II are correct

15. Given below are two statements. One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : H_2S has lower boiling point than H_2O.

Reason (R) : Substances with hydrogen bonding tend to have unusually low boiling points.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(B) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(C) (A) is correct but (R) is not correct.

(D) (A) is not correct but (R) is correct.

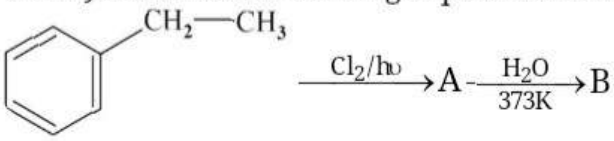
16. Kjeldahl's method can be used for

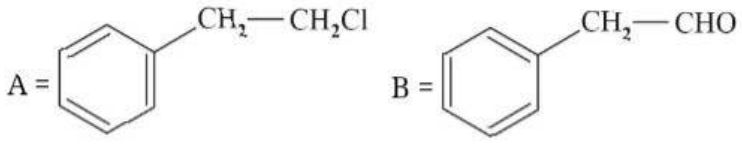
(A) Methanol **(B)** Nitrobenzene **(C)** Pyridine **(D)** Aniline

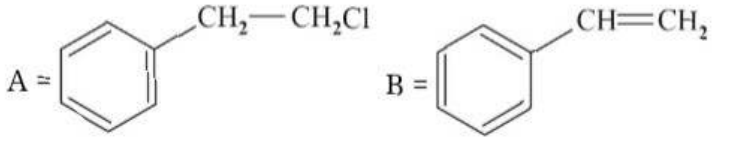
17. Which nitrogenous base is not present in DNA?

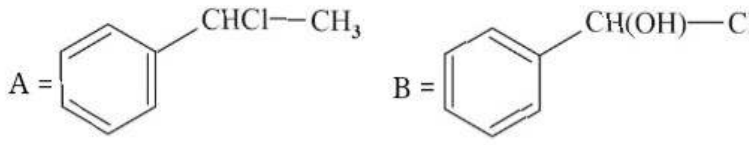
(A) Adenine **(B)** Uracil **(C)** Guanine **(D)** Thymine

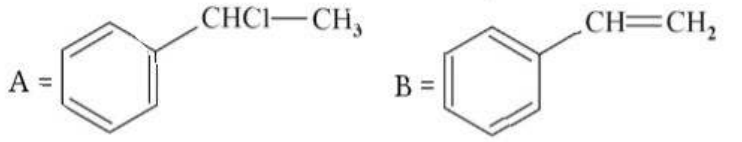
18. Identify A and B in the following sequence of reaction:



(A) 

(B) 

(C) 

(D) 

19. Arrange the bonds in order of increasing ionic character in the molecules.

LiF, KF, O_2, CO_2 and Na_2O

(A) $O_2 < CO_2 < Na_2O < KF < LiF$ **(B)** $LiF < KF < Na_2O < CO_2 < O_2$

(C) $O_2 < CO_2 < Na_2O < LiF < KF$ **(D)** $Na_2O < O_2 < CO_2 < LiF < KF$

20. In H-atom, if 'x' is the radius of first Bohr orbit, de Broglie wavelength of an electron in 5th orbit is:

(A) $6\pi x$ **(B)** $10\pi x$ **(C)** $10x$ **(D)** $5\pi x$

SPACE FOR ROUGH WORK

SECTION-2

Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted.
The answer to each question should be **rounded off to the nearest integer**.

1. The number of molecules/ion/s having trigonal planar shape is _____.
NH_3, CO_3^{2-}, BF_3, CO_2, SO_3, $COCl_2$, BrF_5, $[PtCl_4]^{2-}$, NO_3^-, SO_4^{2-}

2. Number of optical isomers possible for 4-Bromopentan-2-ol.

3. Consider the following reaction $4FeCl_3 + 3K_4[Fe(CN)_6] \rightarrow \underset{\text{Ferriferrocyanate}}{Fe_4[Fe(CN)_6]_3} + 12KCl$.

If 60 mmol of $FeCl_3$ is mixed with 48 mmol of $K_4[Fe(CN)_6]_3$, then the amount of $Fe_4[Fe(CN)_6]_3$ formed is _____ mmol (Nearest integer).

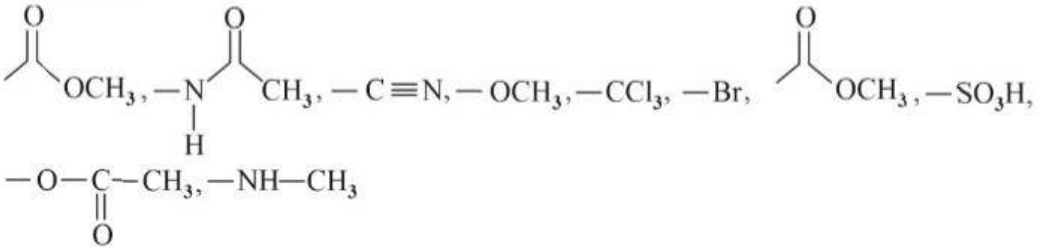
4. Among the following oxides, number of oxides having only acidic nature is _____.
Cl_2O_7, CO, PbO_2, N_2O, NO, Al_2O_3, SiO_2, N_2O_5, Na_2O, Mn_2O_7, CrO_3, SO_2

5. The cell Pt, H_2 (1atm) | H^+ (pH = ?) ||I^- (2M) |$AgI_{(s)}$, $Ag_{(s)}$ |Pt has emf $E_{298K} = 0$. The standard electrode potentials for the reaction $AgI + e^- \rightarrow Ag + I^-$ is –0.162 volt. pH of the solution is _____.
[Given : 2.303 RT/F = 0.06 V, log (2) = 0.3]

6. The lowest oxidation state of an atom in a compound A_3B is –3. The group number of that element is_____.

7. A sample of U^{238} (Half-life = 4.5×10^9 years) ore is found to contain 11.9 g U^{238} and 30.9 g of Pb^{206}. If the age of the ore is $x\times10^9$ years, then value of x is _____.

8. Total number of activating groups in aromatic electrophilic substitution reaction among the following is_____.



9. The number of yellow-coloured salts, among the following is _____.

(a)	$SnSO_4$	**(b)**	$BaSO_4$	**(c)**	AgI	**(d)**	PbI_2
(e)	$PbCrO_4$	**(f)**	Ag_2CrO_4	**(g)**	CdS	**(h)**	HgS
(i)	SnS_2	**(j)**	As_2S_3	**(k)**	$Fe(SCN)_3$	**(l)**	$NaMnO_4$
(m)	CuO						

10. K_a for CH_3COOH is 1.6×10^{-5} and K_b for NH_4OH is 1.6×10^{-5}. The degree of hydrolysis of 0.05 M CH_3COONH_4 is $x\times10^{-1}$. Value of x is _____. (Nearest integer)

SPACE FOR ROUGH WORK

PART III: MATHEMATICS	MARKS: 100

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Let $S=\{x\in R:(3+2\sqrt{2})^x+(3-2\sqrt{2})^x=34\}$, then number of elements in S is:

(A) 0 **(B)** 4 **(C)** 1 **(D)** 2

2. If $3f(x)+2f\left(\frac{1}{x}\right)=\frac{1}{x^2}+1\ \forall\ x\neq 0$ and $y=5x^2f(x)$ then y is strictly decreasing in:

(A) $\left(-\frac{1}{2},0\right)\cup\left(\frac{1}{2},\infty\right)$ **(B)** $\left(-\infty,-\frac{1}{2}\right)\cup\left(0,\frac{1}{2}\right)$

(C) $\left(-\infty,-\frac{1}{2}\right)\cup\left(\frac{1}{2},\infty\right)$ **(D)** $\left(-\frac{1}{2},\frac{1}{2}\right)$

3. Let 5, a, b, c be in AP and 1, $a-6$, $b-4$, $c+10$ be in G.P., then sum of first 5 terms of the GP is:

(A) 242 **(B)** 121 **(C)** 363 **(D)** 726

4. If the area enclosed by the curves $xy+4y=16$ and $x+y=6$ is $(a-b\ln 2)$ square units, then value of $a+b$ is:

(A) 2 **(B)** 60 **(C)** 62 **(D)** 32

5. There are two urns U_1 and U_2. U_1 contains n white and 3 black balls whereas U_2 contains 5 white and 4 black balls. One of the urns is selected at random and a ball is drawn from it. If the drawn ball is white then probability that it is drawn from urn U_1 is $\frac{9}{17}$, then $n=$

(A) 4 **(B)** 5 **(C)** 6 **(D)** 9

6. Let $f:R\to R$ and $g:R\to R$ be such that $f(x)=\begin{cases}\ln x, & x>0\\ e^{-x}, & x\le 0\end{cases}$ and $g(x)=\begin{cases}x, & x\ge 0\\ e^{x}, & x<0\end{cases}$, then $fog:R\to R$ is:

(A) Neither one-one nor onto **(B)** Both one-one and onto

(C) Onto but not one-one **(D)** One-one but not onto

7. If n is the number of ways in which 7 people can stay in 5 indistinguishable rooms of a hotel, when no room remains empty, then n is equal to:

(A) 135 **(B)** 240 **(C)** 140 **(D)** 90

8. Let $f:R\to R$ be defined as $f(x)=\begin{cases}\frac{ax\sin x-(b-1)e^{\sin x}}{x^2}, & x<0\\ 2x^2+cx+1, & 0\le x\le 1.\\ x+2, & x>1\end{cases}$

If f is continuous $\forall\ x\in R$ and m is the number of points where $f(x)$ is NOT differentiable then $m+a+b+c=$

(A) 2 **(B)** 4 **(C)** 1 **(D)** 3

9. The value of integral $\int_0^{\pi/6} \frac{x\,dx}{\sin^4(3x)+\cos^4(3x)}$ equals to:

(A) $\frac{\sqrt{2}\,\pi^2}{36}$ **(B)** $\frac{\sqrt{2}\,\pi^2}{72}$ **(C)** $\frac{\sqrt{2}\,\pi^2}{64}$ **(D)** $\frac{\sqrt{2}\,\pi^2}{18}$

10. If M is a 3×3 matrix such that $M^2 = O$, then $\det((I+M)^{50} - 50M) =$
(Where I is an identity matrix of order 3×3 and O is null matrix)

(A) 3 **(B)** 50 **(C)** 0 **(D)** 1

11. If the system of equations $\left.\begin{aligned} x+2y-z&=4\\ \alpha x+y+3z&=-2\\ 2x-y+\beta z&=1\end{aligned}\right\}$ has infinitely many solutions, then $48\alpha\beta$ is equal to:

(A) 289 **(B)** 179 **(C)** 279 **(D)** 286

12. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} + y(\ln x) = (1+(\ln x))e^x$, $y(1) = e$ then $\left[y\left(-\frac{1}{2}\right)\right]^2 =$

(A) $\frac{1}{e}$ **(B)** e **(C)** $\frac{2}{e}$ **(D)** $2e$

13. Let $\vec{a} = 2\hat{i}-\hat{j}+\hat{k}$, $\vec{b} = \hat{i}+2\hat{j}-3\hat{k}$ and $\vec{c} = (((\vec{a}\times\vec{b})\times\hat{i})\times\hat{j})\times\hat{k}$ then $\vec{c}\cdot(2\hat{i}+\hat{j}-\hat{k}) =$

(A) –7 **(B)** –12 **(C)** –10 **(D)** –14

14. Let $C: x^2+y^2 = 4$ and $C': x^2+y^2-8\lambda x+9 = 0$ be two circles. If the set of all values of λ, so that the circles C and C' intersect at two distinct points, is $R - [a, b]$, then the point $(48a + 40, 80b - 60)$ lies on the curve

(A) $y^2 = 24x$ **(B)** $7x^2 - y^2 = 10$

(C) $(5x)^2 + y^2 = 50$ **(D)** $5x^2 + y^2 = 50$

15. Let $\frac{x^2}{a^2}+\frac{y^2}{b^2} = 1$, $a > b$ be an ellipse, whose eccentricity is $\frac{\sqrt{3}}{2}$ and length of latus rectum is 5, then square of eccentricity of conjugate hyperbola of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2} = 1$ is:

(A) 3 **(B)** 7 **(C)** 5 **(D)** $\frac{5}{2}$

16. Let $S = \left\{z \in C : |z-3| = 1 \text{ and } (1+5\sqrt{3}i)z+(1-5\sqrt{3}i)\bar{z} = 0\right\}$.
Let $z_1, z_2 \in S$ be such that $|z_1| = \max_{z\in S}|z|$ and $|z_2| = \min_{z\in S}|z|$ then $|z_1 \cdot z_2| =$

(A) 8 **(B)** 6 **(C)** 7 **(D)** 5

17. If $\tan A = (x^{-3}+x^{-2}+x^{-1})^{1/2}$, $\tan B = \frac{1}{\sqrt{x}\sqrt{x^2+x+1}}$ and $\tan C = \frac{\sqrt{x}}{\sqrt{x^2+x+1}}$. where, $0 < A, B, C < \frac{\pi}{2}$ then, $(A - B)$ is equal to:

(A) $\frac{\pi}{2}-C$ **(B)** C **(C)** $\pi-C$ **(D)** $\frac{\pi}{2}+C$

18. Let the median and mean deviation about the median of 7 observations. 170, 125, 230, 190, 210, a, b be 210 and $\frac{241}{7}$ respectively, then mean of all seven data is:

(A) 202 **(B)** 203 **(C)** 204 **(D)** 201

19. If the shortest distance between the lines $\frac{x-\lambda}{-2}=\frac{y-2}{1}=\frac{z-1}{1}$ and $\frac{x-\sqrt{3}}{1}=\frac{y-1}{-2}=\frac{z-2}{1}$ is 2, then sum of squares of all possible values of λ is:

(A) 27 **(B)** 24 **(C)** 36 **(D)** 30

20. For $0<\theta<\frac{\pi}{2}$, if the eccentricity of the hyperbola $x^2-y^2\sec^2\theta=5$ is $\sqrt{7}$ times of the eccentricity of the ellipse $x^2\sec^2\theta+y^2=5$, then the value of θ is:

(A) $\frac{\pi}{4}$ **(B)** $\frac{5\pi}{12}$ **(C)** $\frac{\pi}{6}$ **(D)** $\frac{\pi}{3}$

SPACE FOR ROUGH WORK

SECTION-2

Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted.
The answer to each question should be **rounded off to the nearest integer**.

1. Let 2, 7, 12, 17, ….., 404 terms and 3, 5, 7, 9….up to 200 terms be two arithmetic progressions, then the sum of the common terms in them, is equal to _____.

2. Let the line $L: x+\sqrt{3}y = \alpha$ passes through the point of the intersection P (in the first quadrant) of the circle $x^2+y^2 = 4$ and the parabola $y^2 = 3x$. Let the line L touch two circles C_1 and C_2 of equal radius 2. If the centres Q_1 and Q_2 of the circles C_1 and C_2 lie on the x-axis, then the square of the area of the triangle PQ_1Q_2 is equal to _____.

3. If $\int_{-\pi/4}^{\pi/4} \frac{4\sqrt{2}\cos^2 x}{(1+(2024)^x)(\sin^4 x+\cos^4 x)}dx = \alpha\pi+\beta\log_e(\sqrt{2}-1)$ where α, β are integers, then $\alpha^2+\beta^2$ equals _____.

4. Let $A = \{1, 2, 3, \ldots\ldots,20\}$. Let R_1 and R_2 be two relation on A such that:
$R_1 = \{(a, b) : b \text{ is divisible by } a\}$
$R_2 = \{(a, b) : a < b\}$
Then, number of elements in $R_2 - R_1$ is equal to _____.

5. Let the line of the shortest distance between the lines $L_1: \vec{r} = \hat{i}+\lambda(\hat{j}-\hat{k})$ and $L_2: \vec{r} = \hat{j}+\hat{k}+\mu(\hat{i}-\hat{j})$ intersect L_1 and L_2 at P and Q respectively. If (α, β, γ) be the mid-point of the line segment PQ then $6(\alpha+\beta+\gamma)$ is equal to _____.

6. If the coefficient of x^{30} in the expansion of $\left(1+\frac{1}{x^3}\right)^6(1+x^2)^7(1-x^3)^8$, $x \neq 0$ is α, then $|\alpha|$ equals _____.

7. Let $P = \{z \in C : |z-3-2i| \leq 2\}$ and $Q = \{z \in C : z(1-i)+\bar{z}(1+i) \geq 8\}$. Let in $P \cap Q$, $|z+2+3i|$ be maximum and minimum at z_1 and z_2 respectively. If $|z_1|^2+2|z_2|^2 = \alpha+\beta\sqrt{2}$, where α, β are integers, then $\alpha+\beta$ equals to _____.

8. The number of elements in the set $S = \{(x, y, z) : x, y, z \in Z, 2x+2y+3z = 32, x, y, z \geq 0\}$ equals to_____.

9. If $x = x(t)$ be the solution of the differential equation $(t+1)dx = (2x+t^4+4t^3+6t^2+4t+1)dt$, $x(0) = \frac{1}{2}$ then $x(1)$ equals to _____.

10. Let $\{x\}$ denote the fractional part of x and $f(x) = \frac{\cos^{-1}(1-\{x\}^2)\sin^{-1}(\{x\}-\{x\}^2)}{\{x\}-\{x\}^3}$, $x \neq 0$. If L and R respectively denote the left-hand limit and right hand limit of $f(x)$ at $x = 0$, then $\frac{32}{\pi^2}(L^2+R^2)$ is equal to_____.

SPACE FOR ROUGH WORK

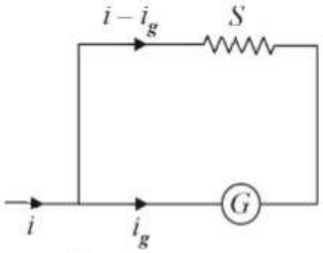
End of 2024 | Mock JEE Main - 10 (CBT) |

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PHYSICS

SECTION-1

1.(B) $S(i-i_g)=i_gG$



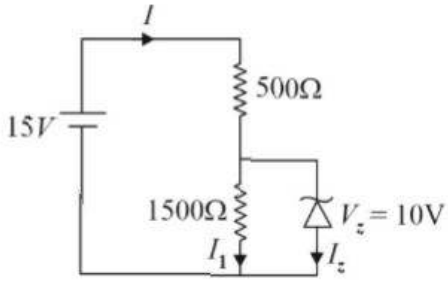
$$S=\frac{i_gG}{i-i_g}=\frac{0.2}{10-\frac{1}{100}}=\frac{0.2}{10}=0.02\Omega$$

2.(D) $I=\frac{15-10}{500}=\frac{1}{100}A$

$I_1=\frac{10-0}{1500}=\frac{1}{150}A$

$I_z=\frac{1}{100}-\frac{1}{150}=\frac{1}{300}A$

$P=V_zI_z=10\times\frac{1}{300}=\frac{1}{30}W=\frac{1000}{30}mW=33.3\times10^{-3}W$



3.(C) $\omega=\frac{1}{\sqrt{LC}}\Rightarrow\omega^2=\frac{1}{LC}\Rightarrow L=\frac{1}{c\omega^2}=\frac{1}{10\times10^{-6}\times10^6}H=\frac{1000}{10}=100mH$

4.(A) $E=h\upsilon\Rightarrow h=\frac{E}{\upsilon}$

$[h]=\frac{ML^2T^{-2}}{T^{-1}}=[ML^2T^{-1}]$

5.(A) $W'=\frac{3}{5}W\Rightarrow mg'=\frac{3}{5}mg\Rightarrow g'=\frac{3}{5}g$

$\Rightarrow g-\omega^2R=\frac{3}{5}g\Rightarrow\omega^2R=\frac{2}{5}g\Rightarrow\omega=\sqrt{\frac{2g}{5R}}$

6.(D) $\frac{V_{He^+}}{V_H}=\frac{\frac{Z_1}{n_1}}{\frac{Z_2}{n_2}}=\frac{Z_1}{Z_2}\times\frac{n_2}{n_1}=\frac{2}{1}\times\frac{3}{3}=\frac{2}{1}$

7.(B) $a=\frac{mg-\mu mg}{2m}=0.4g\,m/s^2$

8.(A) The maximum tension will be at the lowest point.

$T-mg=\frac{mv^2}{r}\Rightarrow T=\frac{mv^2}{r}+mg=\frac{1.5\times4}{0.5}+1.5\times10=27\,N$

9.(D) $\frac{2+3}{\gamma_{mix}-1}=\frac{2}{\gamma_1-1}+\frac{3}{\gamma_2-1} \Rightarrow \frac{5}{\gamma_{mix}-1}=\frac{2}{\frac{2}{3}}+\frac{3}{\frac{2}{5}} \Rightarrow \gamma_{mix}=\frac{31}{21}$

10.(B) $\cos\phi=\frac{R}{Z}=\frac{R}{\sqrt{R^2+X^2}}=\frac{3}{5}=0.6$

11.(C) $n_1=3\rightarrow n_2=1$

$\Delta E=13.6\times 9\times\left[1-\frac{1}{9}\right]=108.8eV$

12.(B) $W=\frac{nR\Delta T}{1-x}=\frac{1\times R\times 2T_0}{1-\frac{3}{4}}=8RT_0$

13.(C) $\frac{\Delta T}{T}\times 100=\frac{1}{2}\left(\frac{\Delta l}{l}\times 100\right)+\frac{1}{2}\left(\frac{\Delta g}{g}\times 100\right)$

$=\frac{1}{2}\left(\frac{0.05\times 100}{5}\right)+\frac{1}{2}\left(\frac{0.2}{10}\times 100\right)=\frac{1}{2}\times 1\%+\frac{1}{2}(2\%)=1.5\%$

14.(A) $R=2H \Rightarrow \cot\theta=\frac{1}{2};\ \sin\theta=\frac{2}{\sqrt{5}},\ \cos\theta=\frac{1}{\sqrt{5}}$

$\therefore$ Range of projectile $R=\frac{2v^2\sin\theta\cos\theta}{g}=\frac{4v^2}{5g}$

15.(A) $S=\frac{D}{d}(\mu-1)t=\frac{5\lambda D}{d} \therefore \lambda=\frac{(\mu-1)t}{5}=6000$ Å.

16.(C) $\text{LC}=\frac{\text{Pitch}}{\text{Number of division on circular scale}}=\frac{1mm}{200}=\frac{1}{200}\times 10^3\mu m=5\mu m$

17.(A) Conceptual

18.(C) $v^2=v_0^2-2gR(1-\cos\theta)$

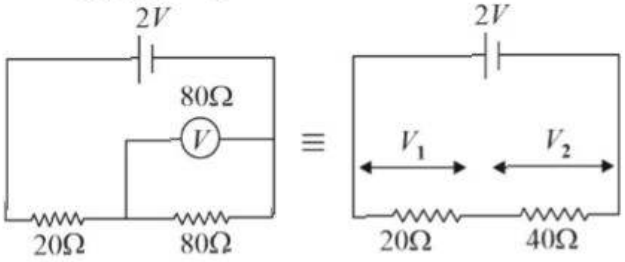
$v^2=7gR+2gR\cos\theta$

$T-mg\cos\theta=\frac{mv^2}{R}$

$T=7mg+3mg\cos\theta$

$v_h=\sqrt{7gr} \Rightarrow T=7mg$

19.(C) $V_2=\frac{40\times 2}{40+20}=\frac{8}{6}=1.33$ volt



20.(D) $U_i = \frac{1}{2}CV^2 + \frac{1}{2}C(2V)^2 = 2.5CV^2 = \frac{5}{2}CV^2$

$V_C = \frac{-CV + 2CV}{2C} = \frac{CV}{2C} = \frac{V}{2}$

$U_f = \frac{1}{2}(C+C)\left(\frac{V}{2}\right)^2 = \frac{1}{2}\times 2C \times \frac{V^2}{4} = \frac{CV^2}{4}$

$\text{Loss} = \left(\frac{5}{2} - \frac{1}{4}\right)CV^2 = \frac{9}{4}CV^2$

SECTION-2

1.(0) $x = \frac{t^3}{3} - 2t^2 + 4t + 3$

$V = t^2 - 4t + 4 = (t-2)^2$

$a = 2t - 4$

Now, $V = 0 \quad \Rightarrow \quad t = 2\,\text{sec}$

So, $a_{t=2} = 2\times 2 - 4 = 0\,m/s^2$

2.(20) $f = \frac{(2n-1)v}{4l} = \frac{(2n-1)}{4l}\sqrt{\frac{RT}{M}}$

$\Rightarrow \quad \frac{\Delta f}{f} = \frac{1}{2}\frac{\Delta T}{T} \quad \Rightarrow \quad \Delta f = \frac{1}{2}\times\frac{1}{250}\times 10000 = \frac{10000}{500} = 20Hz$

3.(363) Continuity Equation

$av_1 = \frac{a}{2}v_2 \quad \Rightarrow \quad v_2 = 2v_1$

Applying Bernoulli's Theorem

$P_1 + \frac{1}{2}\rho v_1^2 + \rho gh = P_2 + \frac{1}{2}\rho v_2^2 + 0$

$P_1 - P_2 = \frac{1}{2}\rho(v_2^2 - v_1^2) - \rho gh$

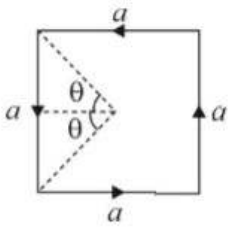
$\Rightarrow \quad 4100 = \frac{1}{2}\times 800[3v_1^2] - 800\times 10\times 1 \quad \Rightarrow \quad v_1 = \frac{\sqrt{363}}{6}$

4.(16) $4a = 4\pi \quad \Rightarrow \quad a = \pi$

$i = \sqrt{2}\pi\ Amp$

$\theta = 45^\circ$

$B = 4\times\frac{\mu_0 i}{4\pi d}[\sin\theta + \sin\theta] = \frac{4\times 4\pi\times 10^{-7}\times 2\pi}{4\pi\times\frac{\pi}{2}} 2\sin 45^\circ = 16\times 10^{-7}T$



5.(300) $W = q\Delta V = 20\times 15 = 300\text{J}$

6.(5) $\frac{f_a}{f_e} = \frac{\left(\frac{\mu_g}{\mu_l} - 1\right)}{\mu_g - 1} = \frac{2}{5} \quad \Rightarrow \quad f_e = \frac{5}{2}f_a = 5cm$

7.(2) Conceptual

8.(2) $i = \dfrac{d\phi/dt}{R} = \dfrac{12t-4}{10} \Rightarrow i(t=2) = \dfrac{20}{10} = 2$

9.(2) **Case I:** $\dfrac{1}{2}mv_1^2 = h(2\nu_0) - h\nu_0$

$\Rightarrow \quad \dfrac{1}{2}mv_1^2 = h\nu_0 \qquad \ldots\ldots$ (i)

Case II : $\dfrac{1}{2}mv_2^2 = h(5\nu_0) - h\nu_0 = 4h\nu_0$

$\Rightarrow \quad \dfrac{1}{2}mv_2^2 = 4h\nu_0 \qquad \ldots\ldots$ (ii)

Divide (ii) by (i)

$$\left(\frac{v_2}{v_1}\right)^2 = 4 \quad \Rightarrow \quad v_2 = 2v_1$$

10.(36) $Q_1 = Q_2 = \dfrac{2.1-0.1}{2} = 1nC = 10^{-9}C$

$$F = \frac{kQ_1Q_2}{r^2} = \frac{9\times10^9\times10^{-9}\times10^{-9}}{0.5\times0.5} = 36\times10^{-9}N$$

CHEMISTRY

SECTION-1

1.(C) $-COOH, -Br \rightarrow$ deactivating group

$-CH_3 \rightarrow$ activating group

$-NH_2$ is strong activating group

2.(D) Number of electrons and principal quantum number (n) are same for all atoms/Ions of isoelectronic series.

3.(D)

(A) $S^{-2} + S^{+4} \rightarrow S^{o}$ comproportionation

(B) $Mn^{+6} \rightarrow Mn^{+7} + Mn^{+4}$ disproportionation

(C) $Mn^{+7} \rightarrow Mn^{+6} + Mn^{+4}$ redox reaction

(D) $Ag^{2+}(aq) + Ag(s) \rightarrow 2Ag^{+}(aq)$ comproportionation

4.(B) For, urea i = 1

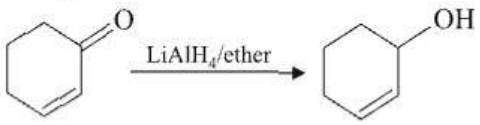
NaCl i = 2

$MgCl_2$ i = 3

5.(A)

$$R-CN \xrightarrow[H_2O]{DiBAL-H} R-CHO$$

$$R-\underset{\underset{O}{\|}}{C}-R \xrightarrow[KOH]{NH_2-NH_2} R-CH_2-R$$



$$R-\underset{\underset{O}{\|}}{C}-CH_2-COOH \xrightarrow{NaBH_4,\ H^+} R-\underset{\underset{OH}{|}}{CH}-CH_2-COOH$$

6.(B) By homolytic bond cleavage $\rightarrow$ Carbon free radicals are formed

Heterolytic bond cleavage $\rightarrow$ Carbocations and carbanions are formed.

7.(D) $MnO_4^- + 8H^{\oplus} + 5e^- \rightarrow Mn^{2+} + 4H_2O$

X = 8. Y = 5, Z = 4, $A = Mn^{2+}$

8.(A) $W = -P_{ext}dV = -2\,[20 - 10] = -20$ atm L

9.(A)

$$\underset{\text{(Ionic)}}{R-X} + KNO_2 \rightarrow \underset{\text{alkyl nitrite}}{R-O-N=O} + KX$$

$$\underset{\text{(Covalent)}}{R-X} + AgNO_2 \rightarrow \underset{\text{(Nitro alkane)}}{R-NO_2} + AgX$$

10.(C) $[Co(NH_3)_6]^{2+} \rightarrow Co^{2+} \rightarrow 3d^7 \rightarrow t_{2g}^5, eg^2 \rightarrow$ paramagnetic (pink colour)

$[Cu(NH_3)_4]^{2+} \rightarrow Cu^{2+} \rightarrow 3d^9 \rightarrow$ paramagnetic (dark blue)

Both complex are paramagnetic and shows colour.

11.(A) Fact

12.(B) Benzene is more reactive than Bromobenzene, so alkylation of benzene will take place instead of Bromobenzene

Aniline and phenol don't undergo Friedel-craft's reactions.

13.(A) $[Fe(NH_3)_4Cl_2]^+$ is heteroleptic, remaining all are homoleptic complex.

14.(A) Phenol and carbolic acid are identical compounds.

15.(C) Order of boiling point (Hydrides of group 16th elements)

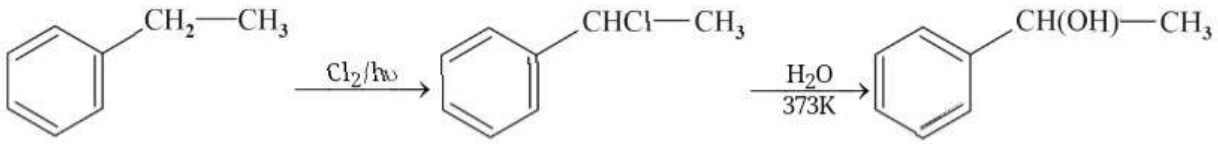
$H_2S < H_2Se < H_2Te < H_2O$ Because of H-bonding

16.(D) Kjeldahl's method is used to determine the nitrogen content in compounds.

Kjeldahl's method not used for compounds containing as nitro, azo groups and for compounds having nitrogen atom in the ring (*e.g.*, pyridine, pyrrole)

17.(B) Uracil base is present in RNA.

18.(C)



19.(C) Ionic character $\propto$ difference in electronegativity of bonded atoms

$O_2 < CO_2 < Na_2O < LiF < KF$

20.(B) $r_1 = x; \quad r_n \alpha \frac{n^2}{1}$

$$\frac{r_1}{r_5} = \frac{1^2}{5^2}$$

$r_5 = 25r_1$ .....(i)

For an electron in Bohr's orbit, $2\pi r = n\lambda$

5th orbit (n = 5)

$2\pi r_5 = 5\lambda \quad \Rightarrow \quad 2\pi \times 25x = 5\lambda \quad \Rightarrow \quad \lambda = 10\pi x$

SECTION-2

1.(5) $BF_3, SO_3, CO_3^{2-}, NO_3^-, COCl_2 \rightarrow$ Trigonal planar (sp^2 hybridisation)

$NH_3 \rightarrow$ Pyramidal (sp^3 hybridisation)

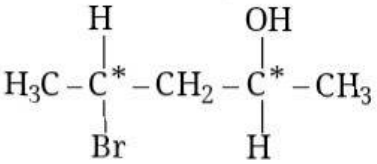
$BrF_5 \rightarrow$ Square pyramidal (sp^3d^2 hybridisation)

$CO_2 \rightarrow$ Linear (sp hybridisation)

$[PtCl_4]^{2-} \rightarrow$ Square planar

$SO_4^{2-} \rightarrow$ Tetrahedral

2.(4) n = 2 (chiral centre)



Total optical isomers = $2^n = 4$

3.(15) $\underset{\text{LR}}{\overset{60}{\downarrow}}$ $4FeCl_3 + 3K_4[Fe(CN)_6] \rightarrow Fe_4[Fe(CN)_6]_3 + 12KCl$ $\underset{\text{ER}}{\overset{48}{\downarrow}}$

Number of mmol of $Fe_4[Fe(CN)_6]_3$ formed $= \frac{60}{4} = 15$

4.(6) $Cl_2O_7, SiO_2, N_2O_5, Mn_2O_7, CrO_3, SO_2 \rightarrow$ Acidic oxides

$CO, N_2O, NO \rightarrow$ Neutral oxides

$Na_2O \rightarrow$ Basic oxide

$PbO_2, Al_2O_3 \rightarrow$ Amphoteric oxide

5.(3) $AgI(s) + \frac{1}{2}H_2(g) \rightarrow Ag(s) + H^+ + I^-$

$$0 = (-0.162) - \frac{0.06}{1}\log\left(\frac{[H^+]\times[I^-]}{1}\right)$$

$$0.162 = -0.06\log(2) - 0.06\log[H^+]$$

$0.18 = 0.06\ (pH) \Rightarrow pH = 3$

6.(15) 15th group elements have minimum -3 Oxidation state.

7.(9) $_{92}U^{238} \rightarrow {}_{82}Pb^{206} + 8\ {}_2He^4 + 6\ {}_{-1}e^0$

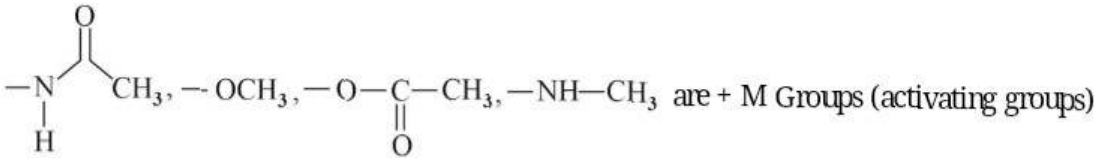
$$n_{Pb} = \frac{30.9}{206} = 0.15 \text{ mol}$$

$$N = (n_U)_{remaining} = \frac{11.9}{238} \Rightarrow 0.05 \text{mol}$$

$$N_0 = (n_U)_{intial} = 0.15 + 0.05 \Rightarrow 0.2 \text{ mol}$$

For first order reaction

$$t = \frac{1}{k}\ln\left(\frac{N_0}{N}\right) = \frac{t_{1/2}}{\ln(2)}\ln\left(\frac{0.2}{0.05}\right) \Rightarrow t = 2t_{1/2} \Rightarrow 9\times10^9 \text{ years}$$

8.(4)  are + M Groups (activating groups)

9.(6) c, d, e, g, i, j $\rightarrow$ coloured species

10.(6) $CH_3COONH_4 \rightarrow$ salt of weak acid and weak base

$$K_h = \frac{K_w}{K_a x K_b}, \qquad K_h = \left(\frac{h}{1-h}\right)^2$$

$$\left(\frac{h}{1-h}\right)^2 = \frac{10^{-14}}{1.6\times1.6\times10^{-10}}$$

$\Rightarrow$ Assume $1 - h \simeq 1 \Rightarrow h = 0.625 \Rightarrow 6.25\times10^{-1}$

MATHEMATICS

SECTION-1

1.(D) $(3+2\sqrt{2})^x+(3-2\sqrt{2})^x=34$

$\Rightarrow \quad t+\frac{1}{t}=34$, Put $(3+2\sqrt{2})^x=t$

$\Rightarrow \quad t^2-34t+1=0 \quad \Rightarrow \quad t=17\pm12\sqrt{2}$

$\Rightarrow \quad (3+2\sqrt{2})^x=17+12\sqrt{2},17-12\sqrt{2}$

$\Rightarrow \quad (3+2\sqrt{2})^x=(3+2\sqrt{2})^2,(3-2\sqrt{2})^2 \quad \Rightarrow \quad x=2,-2$

2.(A) $3f(x)+2f\left(\frac{1}{x}\right)=\frac{1}{x^2}+1$

Replace x by $\frac{1}{x}$

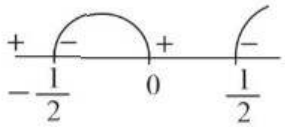
$3f\left(\frac{1}{x}\right)+2f(x)=x^2+1 \qquad \ldots(i)$

On eliminating $f\left(\frac{1}{x}\right)$

$\Rightarrow \quad 5f(x)=\frac{3}{x^2}-2x^2+1 \qquad \ldots(ii)$

$\Rightarrow \quad y=5x^2f(x)=3-2x^4+x^2$

$\Rightarrow \quad \frac{dy}{dx}=-8x^3+2x=-2x(2x+1)(2x-1)$



$\Rightarrow \quad y$ is $\downarrow$ in $\left(-\frac{1}{2},0\right)\cup\left(\frac{1}{2},\infty\right)$

3.(B) Let $\underset{\downarrow}{5}, \ \underset{\downarrow}{a}, \ \underset{\downarrow}{b}, \ \underset{\downarrow}{c}$, d is common difference

$5 \quad 5+d \quad 5+2d \quad 5+3d$

Now, $1, a-6, b-4, c+10$ are in GP (given)

$\Rightarrow \quad 1, d-1, 2d+1, 3d+15$ are in GP

$\Rightarrow \quad \frac{d-1}{1}=\frac{2d+1}{d-1}=\frac{3d+15}{2d+1} \qquad \ldots(i)$

$\Rightarrow \quad d=0, 4$ (on solving)

But $d=0$ does not satisfy (i)

$\therefore \quad d=4 \quad \Rightarrow \quad a=9, b=13, c=17$

$\Rightarrow \quad$ GP : 1, 3, 9, 27, 81

Sum of first 5 terms : 1 + 3 + 9 + 27 + 81 = 121

4.(C) Required area $=\frac{1}{2}(2+8)6-\int_{-2}^{4}\left(\frac{16}{x+4}\right)dx=30-32\ln2$

5.(B) $\underbrace{\overset{n}{(W)}\ \overset{3}{(B)}}_{U_1}\quad \underbrace{\overset{5}{(W)}\ \overset{4}{(B)}}_{U_2}$

Let E_1 = event that urn U_1 is selected

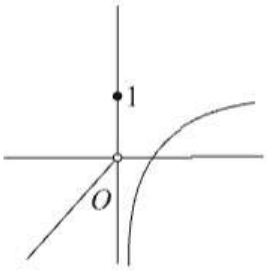
E_2 = event that urn U_2 is selected

A = drawing a white ball from selected urn

Given $P\left(\frac{E_1}{A}\right)=\frac{9}{17}$

$$\Rightarrow \quad \frac{P(E_1).P(A/E_1)}{P(E_1).P(A/E_1)+P(E_2).P(A/E_2)}=\frac{9}{17} \quad \Rightarrow \quad \frac{\frac{1}{2}.\frac{n}{n+3}}{\frac{1}{2}.\frac{n}{n+3}+\frac{1}{2}.\frac{5}{9}}=\frac{9}{17} \quad \Rightarrow \quad n=5$$

6.(C) $f(x)=\begin{cases}\ln x, & x>0\\ e^{-x}, & x\le 0\end{cases}$ and $g(x)=\begin{cases}x, & x\ge 0\\ e^{x}, & x<0\end{cases}$



$$\Rightarrow \quad fog(x)=\begin{cases}x, & x<0\\ 1, & x=0\\ \ln x, & x>0\end{cases}$$

Clearly $fog\ (x)$ not one-one but it is onto.

7.(C)

$\square\ \square\ \square\ \square\ \square$

$1\ \ 1\ \ 1\ \ 1\ \ 3 \rightarrow \quad \frac{7!}{(1!)^4(3!)\cdot 4!}=35$

$1\ \ 1\ \ 1\ \ 2\ \ 2 \rightarrow \frac{7!}{(1!)^3\,(2!)^2.\,3!.2!}=105$

$Total=140$

8.(D) $\because \quad f(x)$ is continuous $\forall \quad x\in R$

$\therefore \quad (LHL)_{x=0}=(RHL)_{x=0}=(V.F)_{x=0}$

$$\Rightarrow \quad \lim_{h\to 0}\left(\frac{ah\sin h-(b-1)e^{-\sin h}}{+h^2}\right)=1$$

For existence of limit $b-1=0$

$\Rightarrow \quad b=1$

Now, $\lim_{h\to 0}\left(\frac{ah\sin h}{h^2}\right)=1$

$\Rightarrow \quad a=1$

Similarly, $(LHL)_{x=1} = (RHL)_{x=1} = (VF)_{x=1}$

$\Rightarrow \quad c + 3 = 3 \quad \Rightarrow \quad c = 0$

Now $f(x)$ becomes $f(x) = \begin{cases} \dfrac{\sin x}{x} & x < 0 \\ 2x^2 + 1 & 0 \le x \le 1 \\ x + 2 & x > 1 \end{cases}$

$\because \quad (LHD)_{x=1} = 4, (RHD)_{x=1} = 1$

At $x = 1$

$f(x)$ is not differentiable and $(LHD)_{x=0} = (RHD)_{x=0} = 0$

At $x = 0$

$f(x)$ is differentiable

Hence, $m = 1$

$m + a + b + c = 1 + 1 + 1 + 0 = 3$

9.(B) $I = \int_0^{\frac{\pi}{6}} \frac{x\,dx}{\sin^4(3x) + \cos^4(3x)}; \qquad I = \frac{\pi}{12}\int_0^{\frac{\pi}{6}} \frac{dx}{\sin^4(3x) + \cos^4(3x)}$

$I = \frac{\pi}{36}\int_0^{\frac{\pi}{2}} \frac{dx}{\sin^4 x + \cos^4 x}; \qquad I = \frac{\pi}{36}\int_0^{\frac{\pi}{2}} \frac{2}{2 - \sin^2 2x}\,dx$

$I = \frac{\pi}{36}\int_0^{\frac{\pi}{2}} \frac{2\sec^2 2x}{2 - \tan^2 2x}\,dx; \qquad I = \frac{\pi}{36}\int_0^{\frac{\pi}{2}} \frac{\sec^2 2x}{2 + \tan^2 2x}\,dx$

$I = \frac{\pi}{18}\int_0^{\frac{\pi}{2}} \frac{\sec^2 x}{2 + \tan^2 x}\,dx$ put $\tan x = t$

$= \frac{\pi}{18}\int_0^{\infty} \frac{dt}{2 + t^2} = \frac{\pi}{18}\cdot\frac{1}{\sqrt{2}}\cdot\left(\tan^{-1}\left(\frac{t}{\sqrt{2}}\right)\right)_0^{\infty} = \frac{\pi}{18\sqrt{2}}\left(\frac{\pi}{2}\right) = \frac{\pi^2}{36\sqrt{2}} = \frac{\pi^2\sqrt{2}}{72}$

10.(D) $|(I + M)^{50} - 50M|$

$= |({}^{50}C_0 I^{50} + {}^{50}C_1 I^{49}.M + {}^{50}C_2 I^{48} M^2 .....) - 50M|$

$= |I + 50M - 50M| = |I| = 1$

($\because \quad M^2 = O$, hence all terms containing higher powers of M. i.e., M^2, M^3, M^4.... will becomes O)

11.(A) $x + 2y - z = 4 \qquad$ ..... (i)

$\alpha x + y + 3z = -2 \qquad$ ..... (ii)

$2x - y + \beta z = 1 \qquad$ ..... (iii)

On eliminating y from (i), (ii), (iii)

$(1 - 2\alpha)x - 7z = 8$ and $(\alpha + 2)x - (\beta + 3)z = -1$

$\because \quad$ System has infinitely many solutions

$\therefore \quad \frac{1-2\alpha}{\alpha+2}=\frac{-7}{\beta+3}=\frac{8}{-1}$

$\Rightarrow \quad \alpha=\frac{-17}{6}, \beta=\frac{-17}{8}$

Now, $48\alpha\beta = 289$

12.(A) $\frac{dy}{dx}+y(\ln x)=(1+\ln x)e^x \rightarrow LDE$

$\Rightarrow \quad I.F.=e^{\int(\ln x)dx}=e^{x\ln x-x}=x^x.e^{-x}$

Now solution of LDE

$y(x^x.e^{-x})=\int(1+\ln x).e^x.x^x.e^{-x}dx+c=\int x^x(1+\ln x)\,dx+c$

$y.x^x.e^{-x}=x^x+c \quad \Rightarrow \quad y=e^x+cx^{-x}.e^x$

$\because \quad y(1)=e \quad \Rightarrow \quad c=0$

$\therefore \quad y=e^x \quad \Rightarrow \quad y\left(-\frac{1}{2}\right)=e^{\frac{-1}{2}} \quad \Rightarrow \quad \left(y\left(-\frac{1}{2}\right)\right)^2=\frac{1}{e}$

13.(A) $\vec{c}=(((\vec{a}\times\vec{b})\times\hat{i})\times\hat{j})\times\hat{k}$

$\because \quad \vec{a}=2\hat{i}-\hat{j}+\hat{k};\ \vec{b}=\hat{i}+2\hat{j}-3\hat{k}$

$\Rightarrow \quad \vec{a}\times\vec{b}=\hat{i}+7\hat{j}+5\hat{k} \quad \Rightarrow \quad \vec{c}=((-7\hat{k}+5\hat{j})\times\hat{j})\times\hat{k}=(7\hat{i})\times\hat{k}=-7\hat{j}$

Now, $\vec{c}.(2\hat{i}+\hat{j}-\hat{k})=-7$

14.(C) $C: x^2+y^2=4; \qquad C': x^2+y^2-8\lambda x+9=0$

$\Rightarrow \quad C_1\equiv(0,0), r_1=2$ and $C_2\equiv(4\lambda,0), r_2=\sqrt{16\lambda^2-9}, \ |\lambda|>\frac{3}{4}$

$\because \quad C$ and C' intersect at two distinct points

$\therefore \quad |r_1-r_2|<C_1C_2<r_1+r_2$

$\Rightarrow \quad |2-\sqrt{16\lambda^2-9}|<|4\lambda|<2+\sqrt{16\lambda^2-9}$

$\Rightarrow \quad \lambda\in R-\left[\frac{-13}{16},\frac{13}{16}\right] \quad \Rightarrow \quad a=\frac{-13}{16}, b=\frac{13}{16}$

Now point $(48a + 40, 80b - 60) = (1, 5)$ which satisfies $(5x)^2+y^2=50$

15.(C) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \quad a>b \quad \Rightarrow \quad e=\sqrt{1-\frac{b^2}{a^2}}=\frac{\sqrt{3}}{2} \quad \Rightarrow \quad \frac{b^2}{a^2}=\frac{1}{4}$

and $LR=\frac{2b^2}{a}=5 \quad \Rightarrow \quad \frac{2a^2}{4a}=5$

$\Rightarrow \quad a=10$ and $b=5$

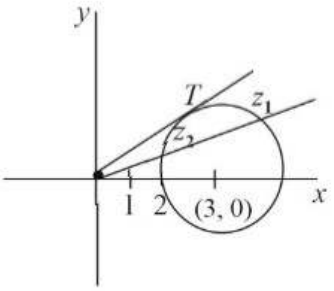
Now eccentricity of hyperbola

$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1 \quad \Rightarrow \quad e_H=\sqrt{1+\frac{b^2}{a^2}}=\sqrt{1+\frac{1}{4}}=\sqrt{\frac{5}{4}}$

Using, $\frac{1}{e_H^2}+\frac{1}{e_{C.H}^2}=1 \quad \Rightarrow \quad e^2_{\text{conjugate hyperbola}}=5$

16.(A) $|z-3|=1$ and $(1+5\sqrt{3}i)z+(1-5\sqrt{3}i)\bar{z}=0$

$\Rightarrow \quad (x-3)^2+y^2=1$ and $x-5\sqrt{3}y=0$



$|z_1.z_2|=|z_1|\,|z_2|$

$=OT^2$

$=(\sqrt{S_1})^2$

$=S_1$

$=(0-3)^2+(0)^2-1=8$

17.(B) $\tan A=\dfrac{\sqrt{1+x+x^2}}{x\sqrt{x}} \qquad \tan B=\dfrac{1}{\sqrt{x}\sqrt{x^2+x+1}}, \ \tan C=\dfrac{\sqrt{x}}{\sqrt{x^2+x+1}}$

$$\tan(A-B)=\frac{\tan A-\tan B}{1+\tan A.\tan B}=\frac{\dfrac{\sqrt{1+x+x^2}}{x\sqrt{x}}-\dfrac{1}{\sqrt{x}\sqrt{x^2+x+1}}}{1+\dfrac{\sqrt{1+x+x^2}}{x\sqrt{x}}.\dfrac{1}{\sqrt{x}\sqrt{x^2+x+1}}}=\frac{\sqrt{x}}{\sqrt{x^2+x+1}}$$

$\tan(A-B)=\tan C \quad \Rightarrow \quad A-B=n\pi+C, (n\in I)$

$A,B,C\in\left(0,\dfrac{\pi}{2}\right) \quad \Rightarrow \quad A-B\in\left(\dfrac{-\pi}{2},\dfrac{\pi}{2}\right)$

$\Rightarrow \quad n=0$

$\Rightarrow \quad A-B=C$

18.(B) Median is 210 $\Rightarrow$ 4th data in ordered form of data is 210

Three data 170, 125, 190 is less than 210. So, a and b > 210

Mean deviation about median

$$=\left(\frac{|210-170|+|210-125|+|210-230|+|210-190|+|210-210|+|210-a|+|210-b|}{7}\right)$$

$$\frac{241}{7}=\frac{40+85+20+20+0+a-210+b-210}{7}$$

$\Rightarrow \quad 241 = a + b - 255$

$\Rightarrow \quad a + b = 496$

Mean of data $=\dfrac{170+125+230+190+210+a+b}{7}=\dfrac{925+496}{7}=\dfrac{1421}{7}=203$

19.(D) Shortest distance $=\left|(\vec{a}_1-\vec{a}_2).\dfrac{(\vec{d}_1\times\vec{d}_2)}{|\vec{d}_1\times\vec{d}_2|}\right|$

$\vec{a}_1-\vec{a}_2=(\lambda,2,1)-(\sqrt{3},1,2)=(\lambda-\sqrt{3},1,-1)$

$$\vec{d}_1\times\vec{d}_2=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\-2&1&1\\1&-2&1\end{vmatrix}=3\hat{i}+3\hat{j}+3\hat{k}$$

$|\vec{d}_1\times\vec{d}_2|=3\sqrt{3}$

$$S.D. = (\lambda - \sqrt{3}, 1, -1).\frac{(\hat{i}+\hat{j}+\hat{k})}{\sqrt{3}}\Bigg|$$

$$2 = \frac{1}{\sqrt{3}}\left|(\lambda - \sqrt{3} + 1 - 1)\right|$$

$$2\sqrt{3} = \left|(\lambda - \sqrt{3}\right|$$

$$\lambda - \sqrt{3} = \pm 2\sqrt{3}$$

$$\lambda = -\sqrt{3},\ 3\sqrt{3}$$

$$\Rightarrow \quad \sum\lambda^2 = (-\sqrt{3})^2 + (3\sqrt{3})^2 = 30$$

20.(C) $H: \frac{x^2}{5} - \frac{y^2}{5\cos^2\theta} = 1; \qquad e_H^2 = 1 + \frac{5\cos^2\theta}{5} = 1 + \cos^2\theta$

$E: \frac{x^2}{5\cos^2\theta} + \frac{y^2}{5} = 1; \quad e_E^2 = 1 - \frac{5\cos^2\theta}{5} = \sin^2\theta$

$e_H = \sqrt{7}e_E \qquad \Rightarrow \qquad e_H^2 = 7e_E^2$

$\Rightarrow \quad 1 + \cos^2\theta = 7\sin^2\theta$

$\Rightarrow \quad \sin^2\theta = \frac{1}{4} \qquad \Rightarrow \qquad \sin\theta = \pm\frac{1}{2}$

$\Rightarrow \quad \theta = \frac{\pi}{6}$ as $\theta \in \left(0, \frac{\pi}{2}\right)$

SECTION-2

1.(8080)

AP(I) : 2, 7, 12, 17, ...404 terms

AP(II) : 3, 5, 7, 9.....200 terms

$\Rightarrow$ 200th term = 401

Now first term in common terms AP, $a = 7$

Common difference d = LCM {5, 2} = 10

For number of common terms $a + (n-1)d < 401$

$\Rightarrow \quad 7 + (n-1)10 < 401 \quad \Rightarrow \quad n < 40.4$

$\Rightarrow$ Number of common terms = 40

Now sum of 40 common terms $= \frac{40}{2}[2(7) + (40-1)10] = 8080$

2.(48) On solving circle $x^2 + y^2 = 4$ and parabola $y^2 = 3x$

Point of intersection in I quadrant $P = (1, \sqrt{3})$

$\because$ L goes from $P = (1, \sqrt{3})$ $\qquad (L: x + \sqrt{3}y = \alpha)$

$\therefore \quad 1 + \sqrt{3}(\sqrt{3}) = \alpha \qquad \Rightarrow \qquad \alpha = 4$

$\Rightarrow$ Line L : $x + \sqrt{3}y = 4$

Let centre of circle C_1 be $(X, 0)$, radius = 2

$\because$ C_1 touches line L

Using $p = r$

$$\left|\frac{x+0-4}{2}\right| = 2 \quad \Rightarrow \quad x = 0, 8$$

$\Rightarrow \quad Q_1 \equiv (0, 0), \; Q_2 = (8, 0)$

Now area of $\Delta PQ_1Q_2 = \frac{1}{2}(8)(\sqrt{3}) = 4\sqrt{3} \quad \Rightarrow \quad \Delta^2 = 48$

3.(5) $I = \int\limits_{\frac{-\pi}{4}}^{\frac{\pi}{4}} \frac{4\sqrt{2}\cos^2 x}{(1+(2024)^x)(\sin^4 x+\cos^4 x)}dx$ ..... (i)

Using rule $\int\limits_a^b f(x)dx = \int\limits_a^b f(a+b-x)dx$

$$I = \int\limits_{\frac{-\pi}{4}}^{\frac{\pi}{4}} \frac{4\sqrt{2}\cos^2 x}{(1+(2024)^{-x})(\sin^4 x+\cos^4 x)}dx$$

$$I = \int\limits_{\frac{-\pi}{4}}^{\frac{\pi}{4}} \frac{4\sqrt{2}(2024)^x.\cos^2 x}{(1+(2024)^x)(\sin^4 x+\cos^4 x)}dx$$

Add (i) + (ii)

$$2I = 4\sqrt{2}\int\limits_{\frac{-\pi}{4}}^{\frac{\pi}{4}} \frac{\cos^2 x}{\sin^4 x+\cos^4 x}dx$$

$$I = 4\sqrt{2}\int\limits_0^{\frac{\pi}{4}} \frac{\cos^2 x}{\sin^4 x+\cos^4 x}dx \quad \Rightarrow \quad I = 4\sqrt{2}\int\limits_0^{\frac{\pi}{4}} \frac{\sec^2 x}{1+\tan^4 x}dx$$

Put $\tan x = t$

$\Rightarrow \quad I = 4\sqrt{2}\int\limits_0^1 \frac{dt}{1+t^4}$

$$\Rightarrow \quad I = 2\sqrt{2}\left[\int\limits_0^1 \frac{1+t^2}{1+t^4}dt + \int\limits_0^1 \frac{1-t^2}{1+t^4}dt\right] = 2\sqrt{2}\left[\int\limits_0^1 \frac{1+\frac{1}{t^2}}{\left(t-\frac{1}{t}\right)^2+2}dt - \int\limits_0^1 \frac{1-\frac{1}{t^2}}{\left(t+\frac{1}{t}\right)^2-2}dt\right]$$

$$= 2\sqrt{2}\left[\frac{1}{\sqrt{2}}\left(\tan^{-1}\left(\frac{\left(t-\frac{1}{t}\right)}{\sqrt{2}}\right)\right)_0^1 - \frac{1}{2\sqrt{2}}\left(\log\left(\frac{t+\frac{1}{t}-\sqrt{2}}{t+\frac{1}{t}+\sqrt{2}}\right)\right)_0^1\right]$$

$$= 2\left(\frac{\pi}{2}\right) - \log\left(\frac{2-\sqrt{2}}{2+\sqrt{2}}\right) = \pi - 2\ln(\sqrt{2}-1) \quad \Rightarrow \quad \alpha^2+\beta^2 = 5$$

$A = \{1, 2, 3, \ldots, 20\}$

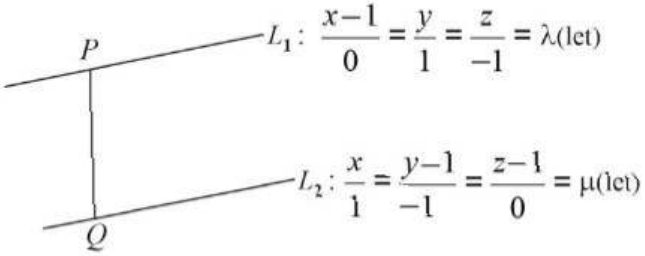
$R_1 = \{(a, b) : b \text{ is divisible by } a\}$

$$\Rightarrow \quad R_1 = \begin{Bmatrix} (1,1), (1,2), \ldots\ldots (1,20) \\ (2,2), (2,4) \ldots\ldots (2,20) \\ \ldots \\ \ldots \\ (20, 20) \end{Bmatrix} \quad \Rightarrow \quad n(R_1) = 66$$

$$R_2 = \{(a, b) : a < b\} = \begin{Bmatrix} (1,2), (1,3), \ldots\ldots (1,20) \\ (2,3), (2,4) \ldots\ldots (2,20) \\ (3,4), (3,3) \ldots\ldots (3,20) \\ \ldots \\ (19,20) \end{Bmatrix} \quad \Rightarrow \quad n(R_2) = 190$$

$$\text{Now, } R_1 \cap R_2 = \begin{Bmatrix} (1,2), (1,3), \ldots\ldots (1,20) \\ (2,4), (2,6) \ldots\ldots (2,20) \\ \ldots \\ (10,20) \end{Bmatrix} \quad \Rightarrow \quad n(R_1 \cap R_2) = 46$$

Now $n(R_2 - R_1) = n(R_2) - n(R_1 \cap R_2) = 190 - 46 = 144$



Let P on L_1 be $(1, \lambda, -\lambda)$ and Q on L_2 be $(\mu, -\mu + 1, 1)$

$\Rightarrow \quad dr's \text{ of } PQ = \mu - 1, -\mu - \lambda + 1, 1 + \lambda$

$\because \quad PQ \perp L_1 = 0(\mu - 1) + 1(-\mu - \lambda + 1) - 1(1 + \lambda) = 0$

$\Rightarrow \quad -\mu - 2\lambda = 0 \qquad \ldots(i) \text{ and}$

$PQ \perp L_2 = 1(\mu - 1) - 1(-\mu - \lambda + 1) + 0(1 + \lambda) = 0$

$\Rightarrow \quad 2\mu + \lambda - 2 = 0 \qquad ..(ii)$

On solving (i) and (ii)

$$\lambda = \frac{-2}{3}, \ \mu = \frac{4}{3} \quad \Rightarrow \quad P \equiv \left(1, \frac{-2}{3}, \frac{2}{3}\right); Q \equiv \left(\frac{4}{3}, -\frac{1}{3}, 1\right)$$

$$\text{Now mid point of } PQ \equiv \left(\underset{\alpha}{\underset{\uparrow}{\frac{7}{6}}}, \underset{\beta}{\underset{\uparrow}{\frac{-1}{2}}}, \underset{\gamma}{\underset{\uparrow}{\frac{5}{6}}} \right)$$

$$\Rightarrow \quad \alpha + \beta + \gamma = \frac{9}{6} \quad \Rightarrow \quad 6(\alpha + \beta + \gamma) = 9$$

6.(0) Coefficient of x^{30} in $\left(1+\frac{1}{x^3}\right)^6 (1+x^2)^7 (1-x^3)^8$

Coefficient of x^{30} in $\frac{(1+x^3)^6}{x^{18}}(1+x^2)^7 (1-x^3)^8$

Coefficient of x^{48} in $(1+x^3)^6 (1+x^2)^7 (1-x^3)^8$

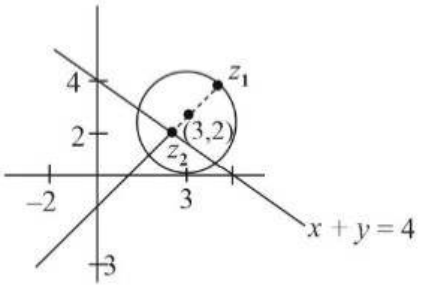
$= {}^8C_8[{}^7C_6 . {}^6C_4 + {}^7C_3 . {}^6C_6] - {}^8C_7[{}^7C_6 . {}^6C_5] + {}^8C_6[{}^7C_6 . {}^6C_6]$

$= [105 + 35] - 8[42] + 196 = 336 - 336 = 0$

7.(44) $P: (x-3)^2 + (y-2)^2 \le 4$

$Q: x + y \ge 4$

z_2 is foot of $\perp$ from $(-2, -3)$ to line $x + y = 4$



$\therefore \quad \frac{x+2}{1} = \frac{y+3}{1} = \frac{-(-2-3-4)}{2}$

$x + 2 = y + 3 = \frac{9}{2} \quad \Rightarrow \quad z_2 \equiv \left(\frac{5}{2}, \frac{3}{2}\right)$

For $|z + 2 + 3i|$ to be maximum z_1 will be diametrically opposite point of the diameter through $(-2, -3)$

For $z_1: \frac{x-3}{\frac{1}{\sqrt{2}}} = \frac{y-2}{\frac{1}{\sqrt{2}}} = 2$

$(x, y) = (3+\sqrt{2}, 2+\sqrt{2}) = z_1$

Now, $|z_1|^2 + 2|z_2|^2 = (3+\sqrt{2})^2 + (2+\sqrt{2})^2 + 2\left(\frac{25}{4} + \frac{9}{4}\right) = 34 + 10\sqrt{2}$

$\alpha = 34, \beta = 10$

8.(57)

z	0	2	4	6	8	10	Total number of solutions
$x + y$	16	13	10	7	4	1	
Number of solutions	17	14	11	8	5	2	17 + 14 + 11 + 8 + 5 + 2 = 57

9.(8) $(t+1)dx = (2x + (t+1)^4)dt$

$\Rightarrow \quad \frac{dx}{dt} - \frac{2x}{t+1} = (t+1)^3 \rightarrow LDE$

I.F. $= e^{-\int \frac{2}{t+1} dt} = e^{-2\ln(t+1)} = \frac{1}{(t+1)^2}$

$x \cdot \frac{1}{(t+1)^2} = \int (t+1)^3 \cdot \frac{1}{(t+1)^2} dt + C$

$\frac{x}{(t+1)^2} = \frac{(t+1)^2}{2} + C \quad \therefore \quad x(0) = \frac{1}{2} \quad \Rightarrow \quad C = 0$

Now, $x = \frac{(t+1)^4}{2} \quad \Rightarrow \quad x(1) = 8$

10.(2) $$L=\lim_{h\to 0}\left(\frac{\cos^{-1}(1-(1-h)^2)\sin^{-1}((1-h)-(1-h)^2)}{(1-h)-(1-h)^3}\right)$$

$$=\lim_{h\to 0}\left(\frac{\cos^{-1}(2h-h^2).\sin^{-1}(h-h^2)}{h^3+2h-3h^2}\right)=\frac{\pi}{2}\lim_{h\to 0}\left(\frac{h-h^2}{h^3+2h-3h^2}\right)$$

$$L=\frac{\pi}{4}$$

$$R=\lim_{h\to 0}\left(\frac{\cos^{-1}(1-h^2).\sin^{-1}(h-h^2)}{h-h^3}\right)=\lim_{h\to 0}\left(\frac{\cos^{-1}(1-h^2)}{h(1-h)(1+h)}.h(1-h)\right)=0$$

$$\Rightarrow \quad \frac{32}{\pi^2}(L^2+R^2)=2$$


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